"PAPER_{0:D3}" -f [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Author: Daniel T. Murphy

"PAPER$_{0:D3}$" -f [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, $n = [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, PAPER_044

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

DPM = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)

Each center is an independent spherical domain containing energy E*DPM* = *?*SCm ■ i■

At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system

Center	h_i	k_i	l_i	State Description
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*pi* = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26* >> *r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase

Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)

Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling

Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i*, *k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

- **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)
- Each center is an independent spherical domain containing energy E*DPM = ?*SCm ■ i■
- At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*pi* = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26* >> *r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

- Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase
- Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)
- Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling
- Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i*, *k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, $n = [int]# "PAPER$_{0:D3}$" -f [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, $n = [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, PAPER_044

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

DPM = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)

Each center is an independent spherical domain containing energy E*DPM* = *?*SCm ■ i■

At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological

Center	h_i	k_i	l_i	State Description
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}$$

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

$$E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})$$

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*p*i = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26 >> r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i = E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase

Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)

Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling

Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i, k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

- **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)
- Each center is an independent spherical domain containing energy E*DPM* = *?*SCm ■ i■

At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster

Center	h_i	k_i	l_i	State Description
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*pi* = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26 >> r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ ri■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase

Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)

Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling

Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

Each center has well-defined quantum numbers (h*i, k*i, l_i) analogous to atomic orbitals

Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)

Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass

Post-inflation level formation follows cosmological cooling sequence

26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, "$\text{PAPER}_{0:D3}$" -f [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, $n = [int]# PAPER #44 ■ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Grok Thread:** b9a29cedc27b45dfa309ea1705721bf0 **Validator:** `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ? **Source Module:** `DPMCosmologyModule.py` (565 lines) **Index Slot:** ■1.6 26-Dimensional Energy Structure, PAPER_044

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1*■*6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

- **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)
- Each center is an independent spherical domain containing energy E*DPM = ?*SCm ■ i■
- At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter

Center	h_i	k_i	l_i	State Description
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*p*i = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26* >> *r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase

Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)

Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling

Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i, k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i,* k*i,* l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + F*p*state). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

DPM = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)

Each center is an independent spherical domain containing energy E*DPM* = ?SCm ■ i■

At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar

Center	h_i	k_i	l_i	State Description
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*pi* = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26* >> *r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase

Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)

Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling

Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i*, *k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value

---

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1■6(U*istate + Fpstate). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

- **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)
- Each center is an independent spherical domain containing energy E*DPM = ?*SCm ■ i■
- At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

---

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*pi* = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26* >> *r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ r*i*■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

- Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase
- Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)
- Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling
- Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i*, *k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57* .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres ■ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h*i*, *k*i, l*i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at t = 0, triggering the universal inflation force F*U(t=0) = F*core + S??1*■*6(U*istate + F*pstate). This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at t = 0. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** ■ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept:

- **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states ■ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense)
- Each center is an independent spherical domain containing energy E*DPM = ?*SCm ■ i■
- At t = 0 (inflation onset), all 26 centers begin simultaneous collapse ? expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

```
h_i = (i-1) \bmod 7 \quad \text{(magnetic quantum number, 0■6)}
```

```
k_i = \lfloor (i-1)/7 \rfloor \quad \text{(angular momentum, increases every 7)}
```

```
l_i = i \quad \text{(radial quantum number)}
```

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed

Center	h_i	k_i	l_i	State Description
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
4	3	0	4	Nucleon pairing
5	4	0	5	Electron shells (K,L)
6	5	0	6	Electron shells (M,N)
7	6	0	7	Valence electrons
8	0	1	8	Van der Waals (h cycle resets)
9	1	1	9	Molecular orbital
10	**2**	**1**	**10**	**Solid matter formation center**
11	**3**	**1**	**11**	**Liquid phase nucleation center**
12	**4**	**1**	**12**	**Gas phase expansion center**
13	**5**	**1**	**13**	**Plasma ionization center**
14	6	1	14	Molecular cluster
15	0	2	15	Cellular (2nd h cycle)
16	1	2	16	Macroscopic matter
17	2	2	17	Centimeter objects
18	3	2	18	Meter-scale
19	4	2	19	Geological
20	**5**	**2**	**20**	**Planetary accretion center**
21	**6**	**2**	**21**	**Stellar formation center**
22	0	3	22	Solar system
23	1	3	23	Interstellar
24	**2**	**3**	**24**	**Galactic structure center**
25	3	3	25	Supercluster
26	**4**	**3**	**26**	**Cosmic web seeding center**

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

```
r_i = 10^{-35} \times 10^{-i/3} \text{ m} = 10^{-35 + i/3} \text{ m}
```

Center	r_i (m)	Comparison
1	2.15■10?■5	~Planck length l_P = 1.616■10?■5

Center	r_i (m)	Comparison
7	4.64■10?■■	~10 ■ Planck
13	1.00■10?■■	sub-nuclear
20	4.64■10?■?	
26	4.64■10?■7	~nuclear scale

2.2 Center Energies

Each center's total energy:

```
E_{\rm center,i} = E_{{\rm DPM},i} \times V_i = \rho_{\rm SCm} \times i^2 \times
\frac{4}{3}\pi r_i^3
```

For center 1: E_center,1 = 10?8 ■ 1 ■ (4/3)p(2.15■10?■5)■ = 10?8 ■ 4.19■10?■■4 = 4.19■10?■■■ J

For center 26: E_center,26 = 10?8 ■ 676 ■ (4/3)p(4.64■10?■7)■ = 6.76■10?6 ■ 4.18■10?7? = 2.83■10?84 J

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

---

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

```
F_U(t=0) = F_{\rm core} + \sum_{i=1}^{26} (U_{i,\rm state} + F_{p,i})
```

where:

F_core = base field force from vacuum [UA]-[SCm] interaction

Ui_state = Universal Inertia at level i (from QuantumLevel26Framework)

F*p*i = thermal/quantum pressure force at level i

Validator confirms: Inflation Force at t=0 ? PASS ?

This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor k*? = 10^10 (K*ETA in DPMCosmologyModule) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

---

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle ?_ij in the pre-inflationary manifold:

```
d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos\theta_{ij}}
```

Adjacent centers (i, j = i+1, ? ■ 2p/26 ■ 13.8■):

d*adjacent* = |*r*{i+1} - r_i| ■ ~1/cos... (small angle limit)

For centers 10,11: d = v((r10■ + r11■) - 2■r10■r11■cos(13.8■))

Distant centers (1 and 26): d*1,26* ■ *r*26 (since r*26 >> r*1)

Validator confirms: Center Separation (adjacent vs distant) ? PASS ?

---

5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

```
S_{\rm mix} = -\sum_{i=1}^{26} p_i \ln p_i
```

where p*i* = *E*{center,i} / E*total is the fractional energy of center i. Since E*center,i ? i■ ■ ri■ = *i*■ ■ *10^(i) (from r*i■ ? 10^i), the energy distribution is strongly weighted toward higher centers ■ most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

Validator confirms: 26-Center Mixing Entropy ? PASS ?

---

6. Level Formation Time Progression

After the Big Bang, the quantum levels form sequentially:

- Lower levels (1■9: nuclear/atomic) form first, during early hot dense phase
- Level 10 (solid matter) forms as universe cools below iron melting temperature (~10,000 K)
- Levels 11■13 (liquid/gas/plasma) form as matter transitions with cooling
- Levels 14■26 form progressively as cosmic structures assemble (stars, galaxies, clusters, web)

Validator confirms: Level Formation Time Progression ? PASS ?

---

7. Scale Factor Evolution

During inflation: `a(t) = \exp(H_{\rm infl} \times t)`, where H*infl from the DPM module uses k*? coupling. At t = 0, the scale factor a(0) = 1 (normalized to the pre-Big Bang manifold scale).

Validator confirms: Scale Factor at t=0 ? PASS ?

Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings:

- Each center has well-defined quantum numbers (h*i*, *k*i, l_i) analogous to atomic orbitals
- Center energies span from ~10?■■■ J (center 1, Planck) to ~10?84 J (center 26)
- Inflation force F_U(t=0) = sum over all 26 centers ■ all tests pass
- Post-inflation level formation follows cosmological cooling sequence
- 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py` *DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57*

UQFF computed: Canonical UQFF buoyancy parameter U*bi = ?■[SSq]■GM/r■ = 5.0e-4■0.57■6.67e-11■M/r■; for solar parameters: U*bi,Sun =

5.7e-4■6.67e-11■1.99e30/(6.96e8)■ = 1.47e+2 m/s■.

Appendix: UQFF Production Framework Reference (v4.75+)

> *Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
> the production physics constants and master equations to enable reproducibility
> against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-}$■ day^{-1}	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k■	1.5	Ug1 DPM-dipole coupling
k■	1.2	Ug2 outer-bubble charge coupling
k■	1.8	Ug3 string-rotation coupling
k■	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
E_react(0)	10■■ J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

```
F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^{4}■igl[\lambda_i
\cdot U_i(r,t) \cdot E_{\mathrm{react}}■igr]
```

Term	Description	Implementation
Ug1	DPM magnetic dipole	`compute_Ug1_SOURCE4` / `compute_Ug1()`
Ug2	Outer-field bubble (charge-reactivity)	`compute_Ug2_SOURCE4` / `compute_Ug2()`
Ug3	Magnetic string rotation	`compute_Ug3_SOURCE4` / `compute_Ug3()`
Ug4	Vacuum concentration (star-BH)	`compute_Ug4_SOURCE4` / `compute_Ug4()`
Ubi	Buoyancy force	`compute_Ubi_SOURCE4` / `compute_Ubi()`
Um	Universal Magnetism (Heaviside-amplified)	`compute_Um_SOURCE4` / `compute_Um()`
−Σλ■·U■·E_react	4th dissipation term (PAPER_420)	`compute_FU_SOURCE4` / full pipeline

4th dissipation term parameters (PAPER_420): λ■=10^{-10}, λ■=10^{-12}, λ■=10^{-11}, λ■=10^{-13} (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

```
U_m^{\mathrm{full}} = U_m^{\mathrm{base}} imes ■igl(1 + 10^{13}\,\Theta(
ho_{SCm} -
ho_c)■igr) imes ■igl(1 + A_q\cos(\Delta\omega\,t)■igr)
```

Symbol	Value	Description
ρ_c	10^{1}■ kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β_i × Ubi	Expanding nebulae, stellar winds
Superconductive	Um × (1+10^{13}·f_H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, *and* `CondensedPhysics2.py`.